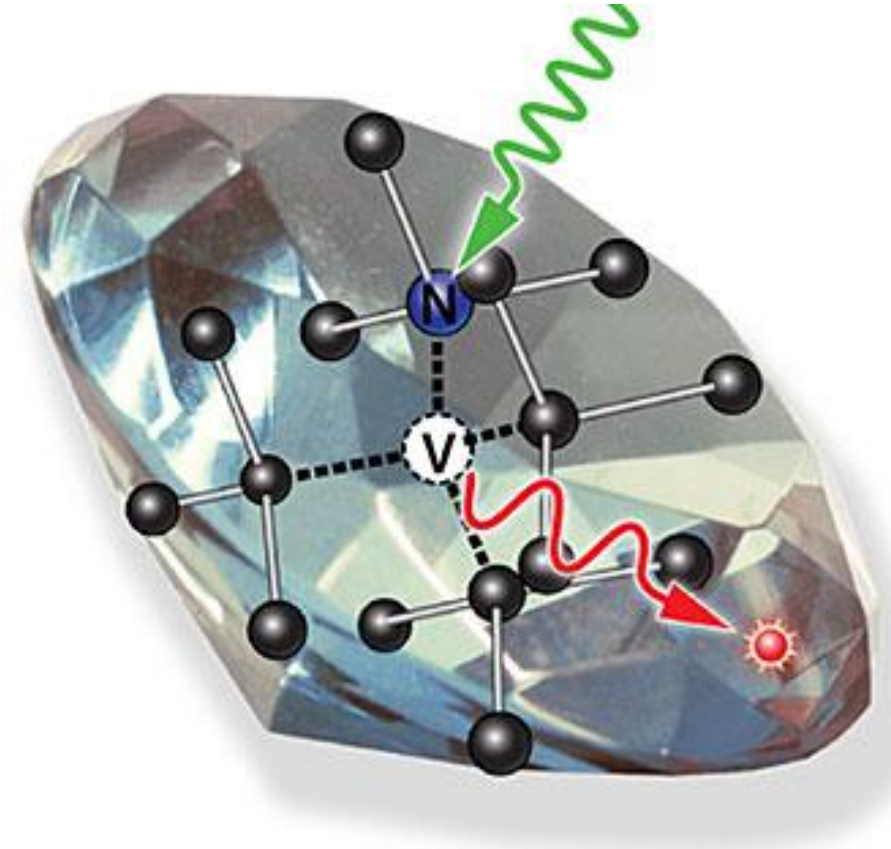


SPIN DYNAMICS SIMULATION IN NV CENTRE DIAMOND



What is NV ensemble ?

Thousands or millions of NV centres clustered together in a crystal diamond lattice.

What is multipartite System ?

- ❑ **Multipartite System**: multiple distinct components or parties.
- ❑ For an NV Centre:
 - **Party 1**: electronic spin triplet
 - **Party 2**: nuclear spin of host ^{14}N atom ($I=1$) or ^{15}N atom ($I=1/2$)
 - **Party 3**: nuclear spins of nearby ^{13}C atoms ($I=1/2$)

Who do we use NV ensemble ?

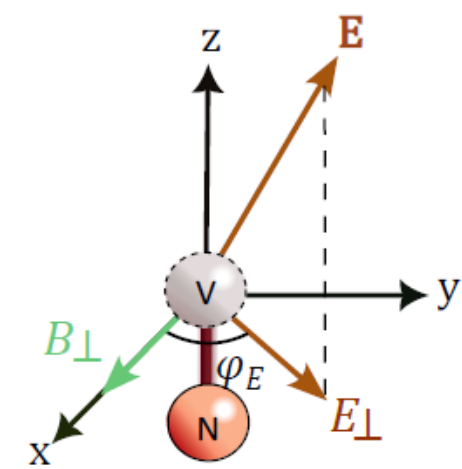
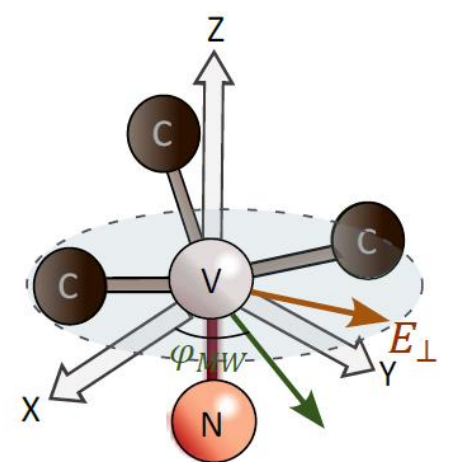
- ❑ Detecting weak fields but over a larger area as compared to single NV centre.
- ❑ Improved SNR

❑ **N** : Number of NV centres

❑ $E_{\min}$: Minimum detectable field strength

$$E_{\min} \propto \frac{1}{\sqrt{N}}$$

$$\text{SNR} \propto \sqrt{N}$$



MF and EF direction

What is Hilbert Space ?

- ❑ Vector Space containing all physically allowable quantum states (state vectors) of a quantum system.
- ❑ Mathematical operations:
 - **Tensor Product** as Lists
 - **Direct Sum** as Tuples

Tensor or Kronecker Product

[Coupled Spin Pair]

- ❑ First spin level : Colour
- ❑ Second spin level : Shape
- ❑ Each state of H_S coexists with each and every state of H_I

Direct Sum

[Pair of electronic energy levels]

- ❑ Ground energy level : Colour
- ❑ Excited energy level : Shape
- ❑ Stacking of separate states in different energy levels.

Two Coupled Spins



S

I

$S = \{ \text{'val': } 1/2, \text{'name': 'S'} \}$

$I = \{ \text{'val': } 1/2, \text{'name': 'I'} \}$

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_I$$

$$\begin{bmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{bmatrix} \otimes \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} = \begin{bmatrix} \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet \end{bmatrix}$$

$s = \text{sos.System}([S, I])$

Two Electronic Levels

$|GS\rangle |ES\rangle$

$GS = \{ \text{'val': } 0, \text{'name': 'GS'} \}$

$ES = \{ \text{'val': } 0, \text{'name': 'ES'} \}$

$$\mathcal{H} = \mathcal{H}_{GS} \oplus \mathcal{H}_{ES}$$

$$\begin{bmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{bmatrix} \oplus \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} = \begin{bmatrix} \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet \end{bmatrix}$$

$a = \text{sos.System}((GS, ES))$

What are Eigen States ?

Stable and naturally existing states under the given Hamiltonian.

❑ From Linear Algebra: $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$

- $\mathbf{M}$: Matrix
- $\mathbf{v}$: Eigen vector
- λ : Eigen value

❑ In the context of NV centre: $\hat{H}|\psi\rangle = E|\psi\rangle$

- $\hat{H}$: Hamiltonian matrix
- $|\psi\rangle$: Eigenstate or Spin State
- E : Energy eigenvalue

❑ Eigen value same $\longrightarrow$ Energy of state same

What is Hamiltonian Operator ?

- Linear operator on a Hilbert Space that describes **total energy** of a physical system.

Ground State Spin Hamiltonian of NV Centre

- $\hat{H}_{nuc}$: Nuclear Spin Hamiltonian
- $\hat{H}_{el}$: Electronic Spin Hamiltonian

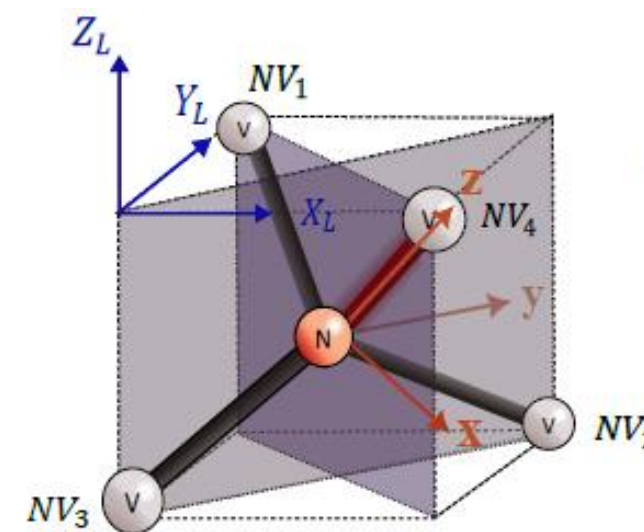
$$\hat{H}_{tot} = \hat{H}_{el} + \hat{H}_{nuc}$$

Static Hamiltonian

- The nuclear spin Hamiltonian is due to the hyperfine interaction between NV electron spin and nitrogen nuclear spin.
- All these Hamiltonians are 9x9 matrices whose eigen values are allowed energy levels of the system.

NV Reference Frame

- Z-axis is along **NV symmetry axis**.
- X-axis is one of the defect's three symmetry planes.
- Fields are decomposed into parallel and transverse components with respect to NV axis.

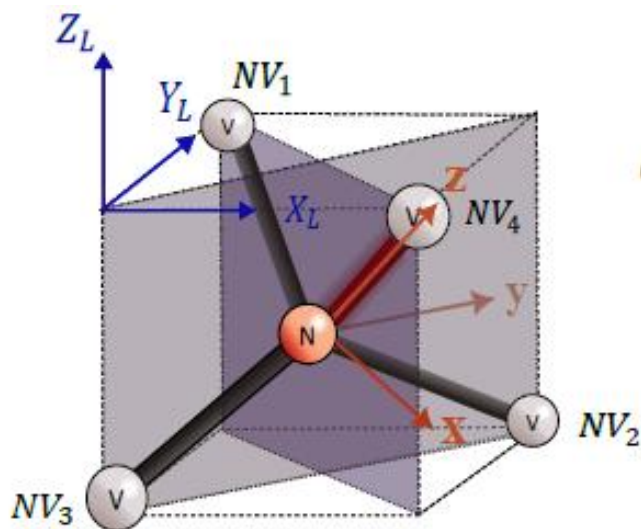


NV Reference Frame and Lab Frame

FRAME TRANSFORMATION

$$T_{NV_1} = \begin{pmatrix} -1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \\ -1/\sqrt{2} & -1/\sqrt{2} & 0 \\ -1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix} \quad T_{NV_2} = \begin{pmatrix} 1/\sqrt{6} & 1/\sqrt{6} & 2/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{3} & 1/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \quad T_{NV_3} = \begin{pmatrix} -1/\sqrt{6} & -1/\sqrt{6} & 2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{3} & -1/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \quad T_{NV_4} = \begin{pmatrix} 1/\sqrt{6} & -1/\sqrt{6} & -2/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}$$

Rotation matrix for Lab frame to NV frame transformation



NV Reference Frame and Lab Frame



4 possible NV orientations in diamond crystal lattice

$$\frac{1}{h}\hat{H}_{el} = (D_{gs} + d_{\parallel}E_z) \left(\hat{S}_z^2 - \frac{2}{3} \right) + \gamma_{NV}\vec{B} \cdot \vec{\hat{S}} - d_{\perp}E_x(\hat{S}_x^2 - \hat{S}_y^2) + d_{\perp}E_y(\hat{S}_x\hat{S}_y + \hat{S}_y\hat{S}_x)$$

Electronic Spin Hamiltonian

Static Hamiltonian
(Time invariant)

- $D_{gs} = 2.87$ GHz : Temperature dependent Zero field splitting
- $\gamma_{NV} = \frac{\mu_B}{h} \cdot g_{NV} = 2.80$ MHz/G : Electronic gyromagnetic ratio
- $d_{\parallel} = 0.35$ Hz cmV⁻¹ : Parallel EF coupling strength
- $d_{\perp} = 17$ Hz cmV⁻¹ : Transverse EF coupling strength
- $\vec{\hat{S}} = (\hat{S}_x, \hat{S}_y, \hat{S}_z)$: Total electronic spin operator

*$d_{\perp}$ is about 50 times larger than $d_{\parallel}$
Larger sensitivity to EF component
perpendicular to NV axis*

GROUND STATE SPIN HAMILTONIAN

$$\frac{1}{h}\hat{H}_{nuc} = A_{\parallel}\hat{S}_z\hat{I}_z + A_{\perp}(\hat{S}_x\hat{I}_x + \hat{S}_y\hat{I}_y) + P\left(\hat{I}_z^2 - \frac{2}{3}\right) + \gamma_N\vec{B} \cdot \vec{\hat{I}}$$

Static Hamiltonian
(Time invariant)

Nuclear Spin Hamiltonian

- ❑ $P = -4.95 \text{ MHz}$: Nuclear quadrupole parameter
- ❑ $\gamma_N = \frac{\mu_N}{h} \cdot g_N = 0.31 \text{ kHz/G}$: Nuclear gyromagnetic ratio
- ❑ $A_{\parallel} = -2.14 \text{ MHz}$: Axial magnetic hyperfine parameter
- ❑ $A_{\perp} = -2.7 \text{ MHz}$: Transverse magnetic hyperfine parameter
- ❑ $\vec{\hat{I}} = (\hat{I}_x, \hat{I}_y, \hat{I}_z)$: Total nuclear spin operator
- ❑ The nuclear quadrupole interaction is only for nuclei with total spin $I \geq 1$

*γ_{NV} is about 9000 times larger than γ_N
So sometimes nuclear term can be ignored for
simplified results*

GROUND STATE SPIN HAMILTONIAN

$$\frac{1}{h} \hat{H}_{int} = \gamma_{NV} \vec{B}_{MW} \cdot \hat{\vec{S}} + \gamma_N \vec{B}_{MW} \cdot \hat{\vec{I}}$$

Dynamic Hamiltonian
(Time varying)

MW Interaction Hamiltonian

Dirac Notation:

$|f\rangle$: Ket

$\langle f|$: Bra

□ δ : Linewidth or Full Width at Half Maximum (FWHM)

□ $\gamma_N = \frac{\mu_N}{h} \cdot g_N = 0.31 \text{ kHz/G}$: Nuclear gyromagnetic ratio

□ $\gamma_{NV} = \frac{\mu_B}{h} \cdot g_{NV} = 2.80 \text{ MHz/G}$: Electronic gyromagnetic ratio

□ $\hat{\vec{S}} = (\hat{S}_x, \hat{S}_y, \hat{S}_z)$: Total electronic spin operator

□ $\hat{\vec{I}} = (\hat{I}_x, \hat{I}_y, \hat{I}_z)$: Total nuclear spin operator

The states $|i\rangle$ and $|f\rangle$ are determined by eigen values of $\hat{H}_{tot}$

□ The interaction Hamiltonian induces transition from **initial eigen state** to **final eigen state**.

Magnetic Dipole Transition Probability

$$T_{i,f} \propto |\langle f | \hat{H}_{int} | i \rangle|^2$$

- * Apply $\hat{H}_{int}$ to $|i\rangle$
- * Inner product the result with $\langle f|$
- * $\langle f|$ is Hermitian of $|f\rangle$
- * Square the modulus of entire term

PAULI SPIN MATRICES

$$\hat{S}_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}; \hat{S}_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}; \hat{S}_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Pauli matrices

```
S_x = 1 / np.sqrt(2) * np.array([[0, 1, 0],
                                  [1, 0, 1],
                                  [0, 1, 0]])
S_y = 1 / np.sqrt(2) * 1j * np.array([[0, -1, 0],
                                       [1, 0, -1],
                                       [0, 1, 0]])
S_z = np.array([[1, 0, 0],
                [0, 0, 0],
                [0, 0, -1]])
SI = np.eye(3)      3x3 identity matrix
```

```
S_zfs = np.dot(S_z, S_z) - 2/3 * SI # Matrix useful for definition of Hamiltonian
```

Since ^{14}N nucleus is a spin-1 ($I=1$) system so $\hat{\vec{I}} = (\hat{I}_x, \hat{I}_y, \hat{I}_z)$ and $\hat{\vec{S}} = (\hat{S}_x, \hat{S}_y, \hat{S}_z)$ are same

HAMILTONIAN CALCULATION

```
def NV_transitionsElevels(B, E):
    # Input: magnetic field and electric field, defined in NV center frame
    # Output: This function diagonalizes the Hamiltonian Hgs and return 9 eignenergies and its vector

    # Hamiltonian
    # Fine and hyperfine terms
    HZFS = D_0 * np.kron(S_zfs, SI) # Zero-field splitting
    HHFPar = Apar * np.kron(S_z, S_z) # Axial hyperfine interaction
    HHFPerp = Aperp * (np.kron(S_x, S_x) + np.kron(S_y, S_y)) # Non-axial hyperfine interaction
    HNucQ = PQ * np.kron(SI, S_zfs) # Nuclear quadrupole interaction

    # Magnetic field coupling terms
    HBE1 = gammaNV * np.kron(B[0]*S_x + B[1]*S_y + B[2]*S_z, SI) # Electric Zeeman coupling
    HBNuc = gammaN * np.kron(SI, B[0]*S_x + B[1]*S_y + B[2]*S_z) # Nuclear Zeeman coupling

    # Electric field coupling terms
    H_elec = (E[2] * d_parallel * np.kron(S_zfs, SI)
              + E[0] * d_transverse * np.kron((np.dot(S_y, S_y) - np.dot(S_x, S_x)), SI)
              + E[1] * d_transverse * np.kron((np.dot(S_x, S_y) + np.dot(S_y, S_x)), SI))

    H_total = HZFS + HBE1 + HBNuc + H_elec + HHFPar + HHFPerp + HNucQ
    E_I, vec_I = np.linalg.eigh(H_total)

    return E_I, vec_I
```

TRANSVERSE EF

$$\hat{H}_{E_{\perp}} = -d_{\perp} E_x (\hat{S}_x^2 - \hat{S}_y^2) + d_{\perp} E_y (\hat{S}_x \hat{S}_y + \hat{S}_y \hat{S}_x)$$

$$\hat{H}_{E_{\perp}} = -d_{\perp} E_x \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} + d_{\perp} E_y \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}$$

$$= d_{\perp} \begin{pmatrix} 0 & 0 & -E_x \\ 0 & 0 & 0 \\ -E_x & 0 & 0 \end{pmatrix} + d_{\perp} \begin{pmatrix} 0 & 0 & -iE_y \\ 0 & 0 & 0 \\ iE_y & 0 & 0 \end{pmatrix}$$

$$= d_{\perp} \begin{pmatrix} 0 & 0 & -E_x - iE_y \\ 0 & 0 & 0 \\ -E_x + iE_y & 0 & 0 \end{pmatrix}$$

In the $\{|+1\rangle, |-1\rangle\}$ subspace:

$$\hat{H}_{E_{\perp}} = d_{\perp} \begin{pmatrix} 0 & -E_x - iE_y \\ -E_x + iE_y & 0 \end{pmatrix}$$

Let $a = -E_x - iE_y$, then:

$$\hat{H}_{E_{\perp}} = d_{\perp} \begin{pmatrix} 0 & a \\ a^* & 0 \end{pmatrix}$$

$$\hat{S}_x^2 - \hat{S}_y^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\hat{S}_x \hat{S}_y + \hat{S}_y \hat{S}_x = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}$$

Eigenvalues of $\begin{pmatrix} 0 & a \\ a^* & 0 \end{pmatrix}$ are $\pm|a|$:

$$|a| = \sqrt{(-E_x)^2 + (-E_y)^2} = \sqrt{E_x^2 + E_y^2} = E_{\perp}$$

Eigenvalues = $\pm d_{\perp} E_{\perp}$

$$\text{Separation} = 2d_{\perp} E_{\perp}$$

Lorentzian Function

$$L(\nu) = \frac{A \cdot (\Gamma/2)^2}{(\nu - \nu_0)^2 + (\Gamma/2)^2}$$

$A \rightarrow$ Peak amplitude
 $\Gamma \rightarrow$ FWHM (linewidth)
 $\nu_0 \rightarrow$ Central frequency

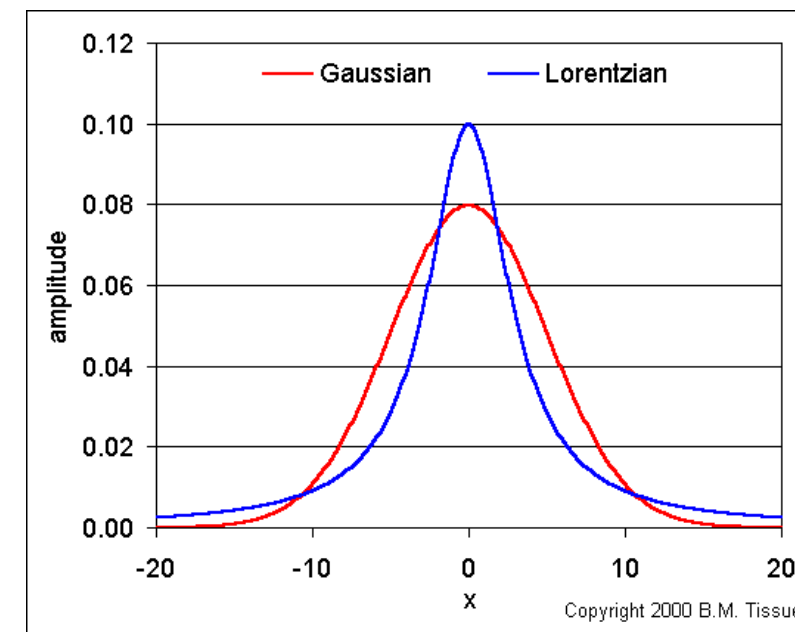
- ❑ This transition has **Lorentzian shape**.
- ❑ Frequency dependent transition amplitude is given as:

$$T_{i,f}(\nu) = \frac{T_{i,f}\delta^2}{4 \left[(\nu - |\nu_f - \nu_i|)^2 + \frac{\delta^2}{4} \right]}$$

Total Transition Strength :

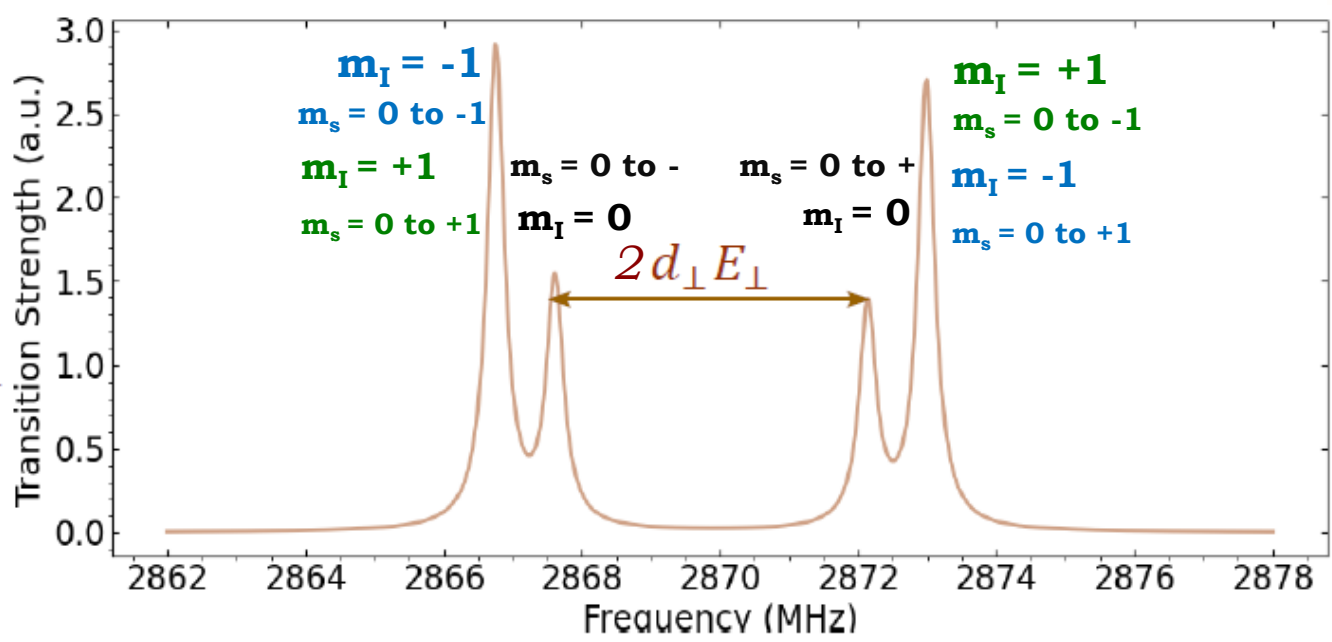
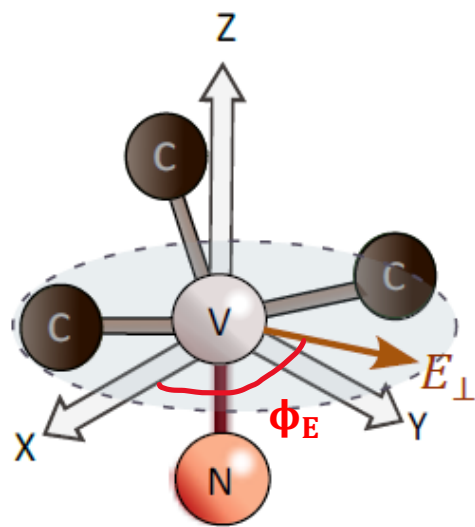
$$T(\nu) = \sum_{i,f} T_{i,f}(\nu)$$

- * Summing over all possible initial and final states
- * T is proportional to **ODMR contrast**



Nature of Gaussian v/s Lorentzian curve

TRANSVERSE EF



Mixing of $|\pm 1\rangle$ states due to transverse EF (MW is unpolarized)

- ❑ Transverse EF mixes the degenerate $|\pm 1\rangle$ states with $m_I = 0$
- ❑ The mixed states are also split by an amount $2d_{\perp} E_{\perp}$ and their spins point in the x-y plane.

$$|-\rangle = \frac{1}{\sqrt{2}}(e^{i\phi_E} |m_s = +1\rangle + |m_s = -1\rangle)$$

$$|+\rangle = \frac{1}{\sqrt{2}}(e^{i\phi_E} |m_s = +1\rangle - |m_s = -1\rangle)$$

AXIAL AND TRANSVERSE EF

$A_{\text{HF}} = -2.14 \text{ MHz}$: Axial hyperfine parameter

$\Pi_{\parallel} = \Pi_z$: Axial/Parallel EF

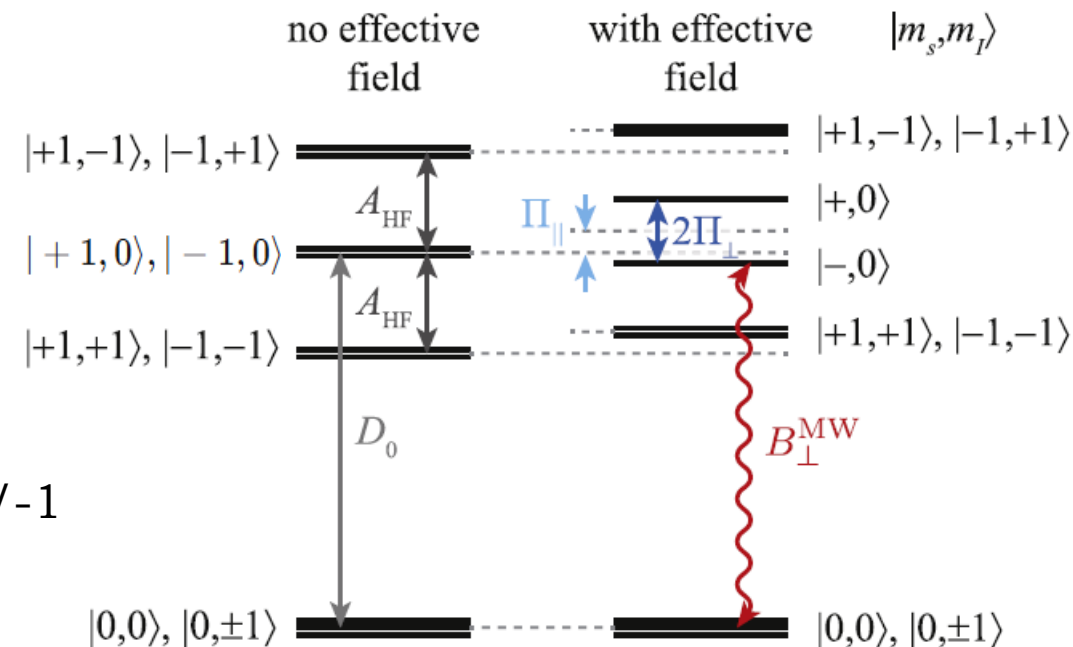
$\Pi_{\perp} = (\Pi_x^2 + \Pi_y^2)^{1/2}$: Transverse/Perpendicular EF

❑ Axial EF causes an equal amount of shift $\Pi_{\parallel}$ in the $m_s = \pm 1$ states wrt $m_s = 0$

❑ The nuclear quadrupole interaction is not taken into account here.

❑ Transverse EF mixes the degenerate $|\pm 1\rangle$ states with $m_I = 0$

❑ The mixed states are also split by an amount $2d_{\perp}\Pi_{\perp}$ and their spins point in the x-y plane.

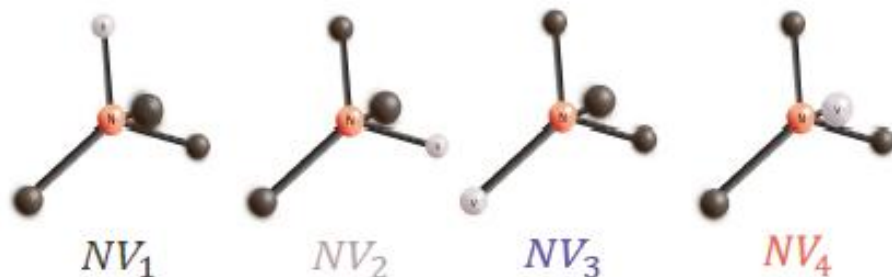


Energy Level Diagram of NV ground state in absence of MF

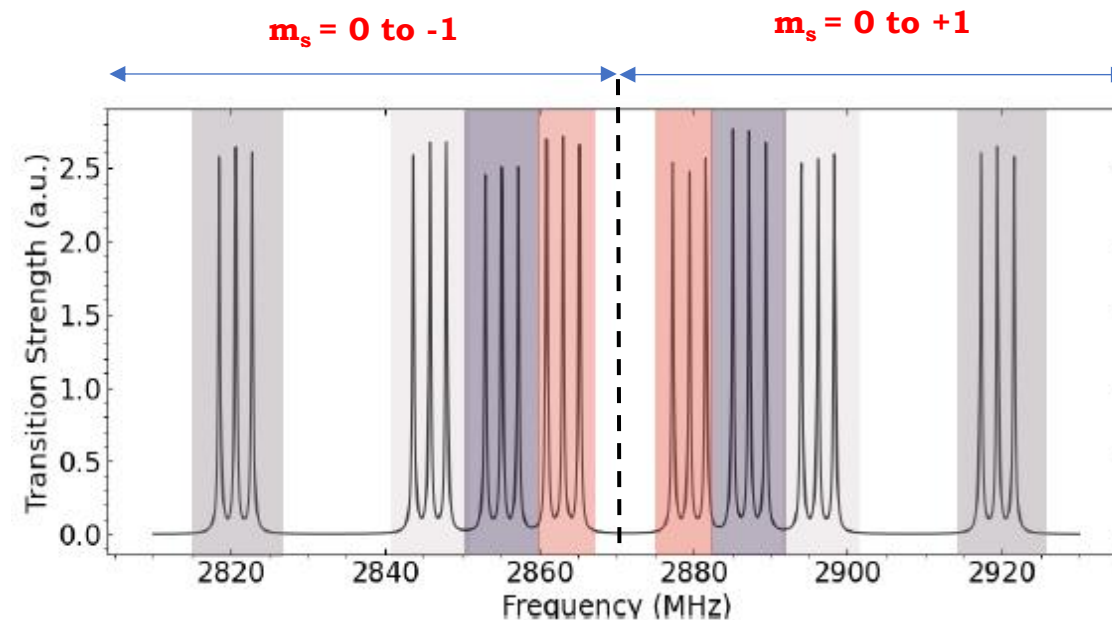
$$|-\rangle = \frac{1}{\sqrt{2}}(e^{i\phi_E}|m_s = +1\rangle + |m_s = -1\rangle)$$

$$|+\rangle = \frac{1}{\sqrt{2}}(e^{i\phi_E}|m_s = +1\rangle - |m_s = -1\rangle)$$

TRANSITION STRENGTH

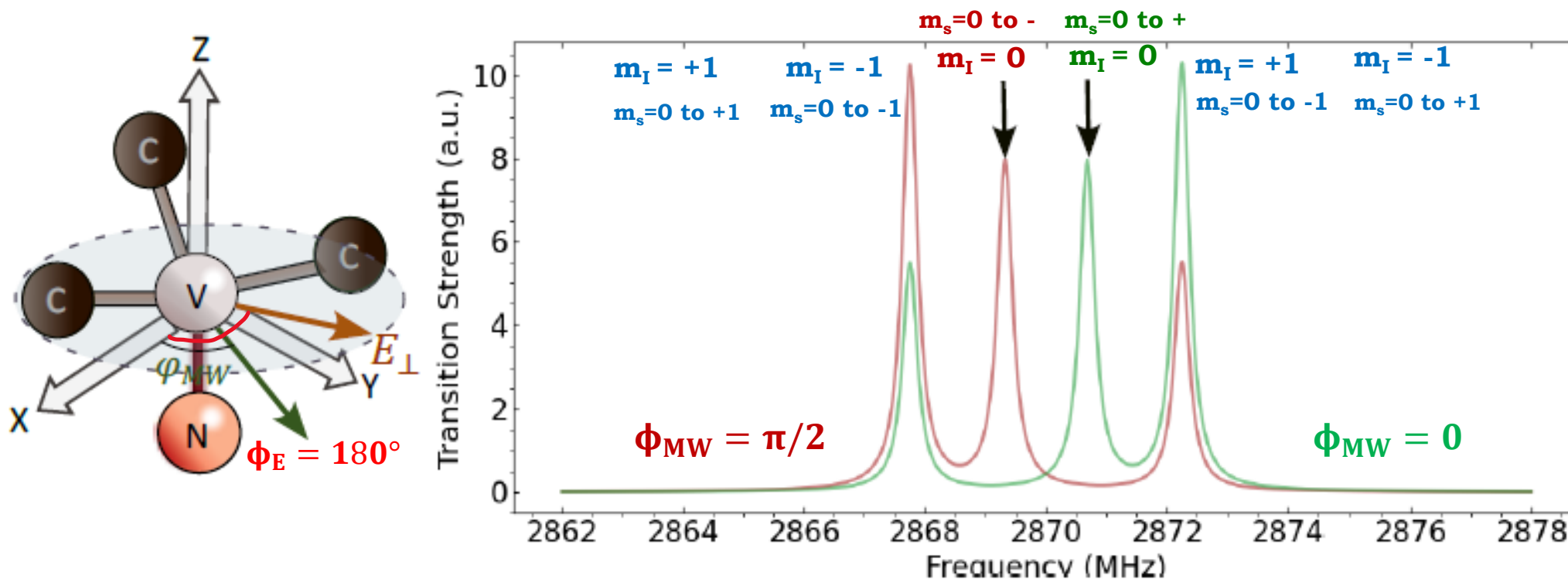


4 possible NV orientations in diamond crystal lattice



Transition strength spectrum at $B = 18G$

4 orientations $\times$ 2 electronic transitions $\times$ 3 hyperfine = 24 peaks



Transition Strength spectrum for two different MW linear polarizations

$m_I = 0$ states (inner hyperfine states, indicated by black arrows) the transition amplitude at lower frequency is completely suppressed for $\phi_{MW} = 0$ and the upper frequency transition has maximal amplitude whereas it is the opposite for $\phi_{MW} = \pi/2$

TRANSITION STRENGTH

$$T_{0,\pm} \propto 1 \mp \cos(2\phi_{MW} + \phi_E)$$

Transition strength from $m_s = 0$ to mixed states $|\pm\rangle$ due to transverse EF and linearly polarized MW signal.

Case 1: $\phi_{MW} = 0, \phi_E = 180^\circ$

Upper peak ($T_{0,+}$):

$$T_{0,+} \propto 1 - \cos(0 + 180^\circ) = 1 - \cos(180^\circ)$$

$$\cos(180^\circ) = -1$$

$$T_{0,+} \propto 1 - (-1) = 2 \quad (\text{Maximal})$$

Lower peak ($T_{0,-}$):

$$T_{0,-} \propto 1 + \cos(0 + 180^\circ) = 1 + \cos(180^\circ)$$

$$T_{0,-} \propto 1 + (-1) = 0 \quad (\text{Suppressed})$$

Case 2: $\phi_{MW} = 90^\circ, \phi_E = 180^\circ$

Upper peak ($T_{0,+}$):

$$T_{0,+} \propto 1 - \cos(180^\circ + 180^\circ) = 1 - \cos(360^\circ)$$

$$\cos(360^\circ) = 1$$

$$T_{0,+} \propto 1 - 1 = 0 \quad (\text{Suppressed})$$

Lower peak ($T_{0,-}$):

$$T_{0,-} \propto 1 + \cos(180^\circ + 180^\circ) = 1 + \cos(360^\circ)$$

$$T_{0,-} \propto 1 + 1 = 2 \quad (\text{Maximal})$$

ODMR CONTRAST

$$T_{0,\pm} \propto 1 \mp \cos(2\phi_{MW} + \phi_E)$$

- ❑ Transition strength from $m_s = 0$ to mixed states $|\pm\rangle$ due to transverse EF and linearly polarized MW signal .
- ❑ Transition Strength $T(\nu)$ is proportional to ODMR contrast in **low magnetic field regime** ($B < 100\text{G}$).
- ❑ Photoluminescence can be measured using **Photodiode**.

$$T(\nu) \uparrow \rightarrow PL \downarrow \text{ (with MW) } \rightarrow \text{ODMR contrast } \uparrow$$

- High transition strength
- large no. of e^- transferred to $|\pm\rangle$ mixed states
- large dip in red fluorescence
- large ODMR contrast.

$$C_{\text{ODMR}} = \frac{Ph_{\text{off}} - Ph_{\text{on}}}{Ph_{\text{off}}}$$

- $Ph_{\text{off}} \rightarrow$ Photoluminescence (off-resonance)
- $Ph_{\text{on}} \rightarrow$ Photoluminescence (on-resonance)

ODMR SPECTRA SIMULATION

Core Libraries

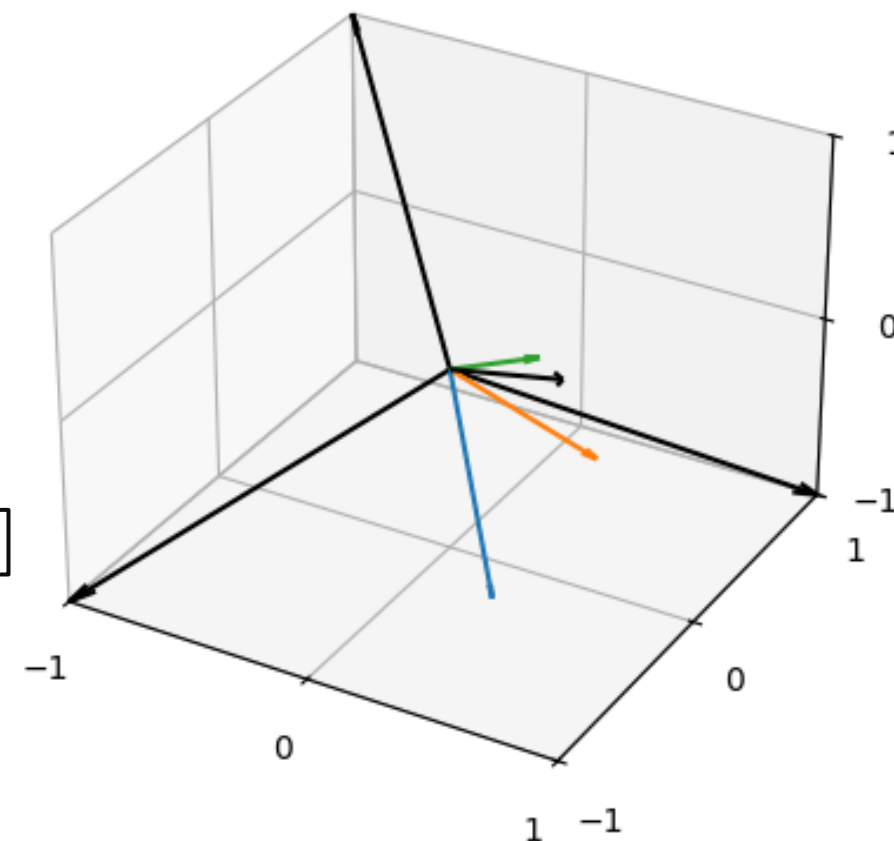
```
import numpy as np : Numerical operations
import matplotlib.pyplot as plt : Plots and Figures
import simulate_TStrength_NV_ensemble as simnv : Custom calculations
```

3D Vector generator : Generating arrow vectors

```
def plot_vector(vec, ax, length=1, color='k'):
    arrow = 0.06 / length : Arrow tip size
    ax.quiver(0, 0, 0, vec[0], vec[1], vec[2],
             length=length, arrow_length_ratio=arrow, color=color)
```

Diamond Lattice 3D Canvas : 3D plot with axes ranging from -1 to +1

```
def plot_config(thetaMW, phiMW, thetaB, phiB, thetaE, phiE):
    fig = plt.figure()
    ax = plt.axes(projection='3d')
    ax.set_xlim(-1, 1)
    ax.set_ylim(-1, 1)
    ax.set_zlim(-1, 1)
    ax.set_xticks([-1, 0, 1])
    ax.set_yticks([-1, 0, 1])
    ax.set_zticks([-1, 0, 1])
```

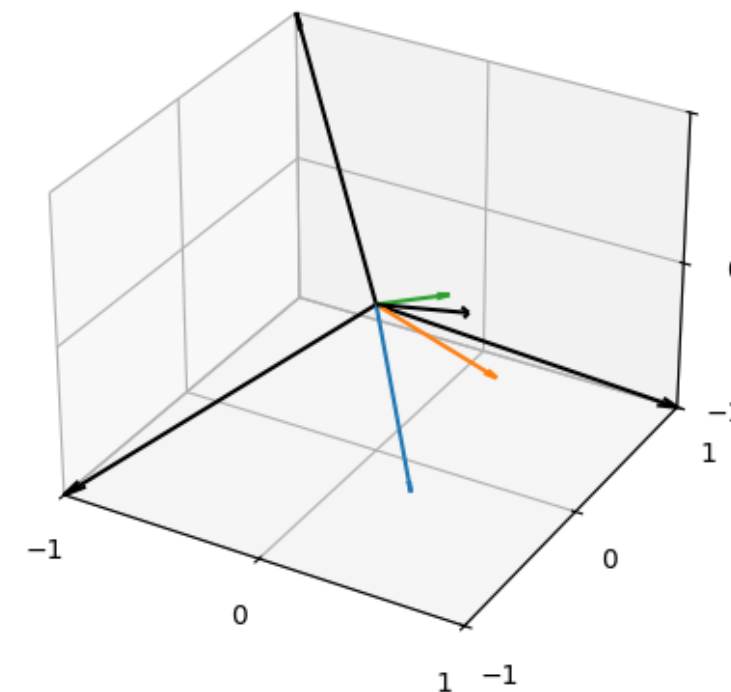


4 NV orientations and EF, MF and MW vectors

ODMR SPECTRA SIMULATION

Crystal Lattice Orientation

```
# add 4 NV orientations
plot_vector([1, -1, 1], ax) # NV1
plot_vector([-1, 1, 1], ax) # NV2
plot_vector([-1, -1, -1], ax) # NV3
plot_vector([1, 1, -1], ax) # NV4
```



MF, EF and MW Vector Plot

4 NV orientations and EF, MF and MW vectors

```
# Plot MW, B and E
plot_vector(simnv.get_vector_cartesian(1, thetaB, phiB), ax, color='C0')
plot_vector(simnv.get_vector_cartesian(1., thetaE, phiE), ax, color='C1', length=2/3.)
plot_vector(simnv.get_vector_cartesian(1., thetaMW, phiMW), ax, color='C2', length=1/3.)
```

ODMR SPECTRA SIMULATION

Transition Strength for Linearly Polarized MW

```
def B_scan(MWfreq, thetaMW, phiMW, B0list, thetaB, phiB, E0, thetaE, phiE, Linewidth):  
    numB = len(B0list)  
    nMW = len(MWfreq)  
    T = np.zeros((numB, nMW))  
  
    for idx, B0 in enumerate(B0list):  
        T[idx, :] = simnv.ESR_NV_VN_ensemble(MWfreq, thetaMW, phiMW, B0, thetaB, phiB, E0, thetaE, phiE, Linewidth)  
  
    return T
```

2D array with Rows = MF values
Columns = MW frequencies

For each MF, transition strength considering all 4 NV orientations and both NV/VN configurations is calculated

Transition Strength for Unpolarized MW

```
def B_scan_unpolarizedMW(MWfreq, thetak, phik, B0list, thetaB, phiB, E0, thetaE, phiE, Linewidth):  
    numB = len(B0list)  
    nMW = len(MWfreq)  
    T = np.zeros((numB, nMW))  
  
    for idx, B0 in enumerate(B0list):  
        T_pol1 = simnv.ESR_NV_VN_ensemble(MWfreq, thetak + np.pi/2., phik, B0, thetaB, phiB, E0, thetaE, phiE, Linewidth)  
        T_pol2 = simnv.ESR_NV_VN_ensemble(MWfreq, np.pi/2., phik + np.pi/2., B0, thetaB, phiB, E0, thetaE, phiE, Linewidth)  
        T[idx, :] = (T_pol1 + T_pol2) / 2.  
  
    return T
```

Statistical average of 2 linearly polarized MW signals gives unpolarized MW

For each MF, transition strength considering all 4 NV orientations and both NV/VN configurations is calculated

ODMR SPECTRA SIMULATION

Data Resolution Limits

```

sdir = '' # path to directory where figures will be saved
save_flag = False # if True, save figures

# MW frequency
nfreq = 2000
freqi = 2860
freqf = 2890

# Linewidth
Linewidth = 1

```

2000 MW frequencies points
centred around resonance point
 $f_o = 2.87$ GHz
FWHM

Magnitude Parameters

Define B and E amplitude

nB = 300

B0 = 80

300 MF values from 0 to 80G

E0 = 5e6

dE = 1e5

$E = 5 \times 10^6$ V/m

$dE = 1 \times 10^5$ V/m

Multi Frame Vector Coordinates

```

# Define MW directions in lab frame
thetaMW = np.pi/2.
phiMW = np.pi/4.

# Define E, B directions in NV1 frame
thetaB = np.pi/2.
phiB = 0.

thetaE = np.pi/2
phiE = np.pi/4.

```

$\theta_{MW} = \frac{\pi}{2}, \quad \phi_{MW} = \frac{\pi}{4}$
 $\theta_B = \frac{\pi}{2}, \quad \phi_B = 0$
 $\theta_E = \frac{\pi}{2}, \quad \phi_E = \frac{\pi}{4}$

Frame Rotation

NV1 frame to Lab Frame conversion

```

# Compute E, B directions in lab frame
thetaB, phiB = simnv.transform_spherical_nv_to_lab_frame(thetaB, phiB)
thetaE, phiE = simnv.transform_spherical_nv_to_lab_frame(thetaE, phiE)

# Plot geometrical configuration
plot_config(thetaMW, phiMW, thetaB, phiB, thetaE, phiE)

```

Run Simulation

```
# Run simulation and plot results
```

```
MWfreq = np.linspace(freqi, freqf, nfreq)
```

```
B0list = np.linspace(0, B0, nB)
```

Creates linearly spaced arrays for MW frequencies and MF strength

```
TSmap = B_scan(MWfreq, thetaMW, phiMW, B0list, thetaB, phiB, E0, thetaE, phiE, Linewidth)
```

```
TSmap_dE = B_scan(MWfreq, thetaMW, phiMW, B0list, thetaB, phiB, E0 + dE, thetaE, phiE, Linewidth)
```

Computes ODMR spectrum map for all (B,MW) combinations at $E=E_0$ and $E=E_0+dE$

Raw ODMR Valley

```
# Figures
```

```
fig_T, ax_T = plt.subplots()
```

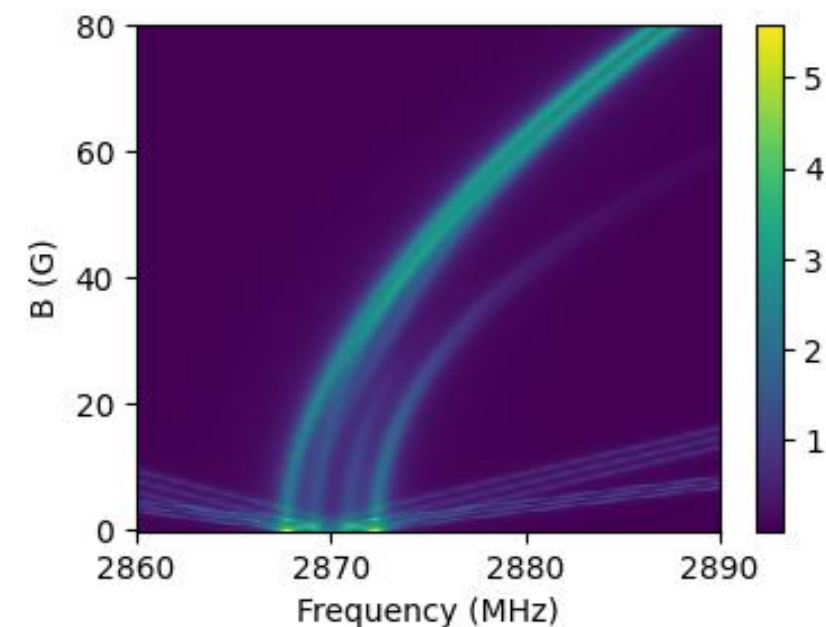
```
im = plt.pcolormesh(MWfreq, B0list, TSmap, shading='auto')
```

```
plt.colorbar(im)
```

```
plt.xlabel('Frequency (MHz)')
```

```
plt.ylabel('B (G)')
```

Shows heat map of transition strength as a function of MF and MW frequency

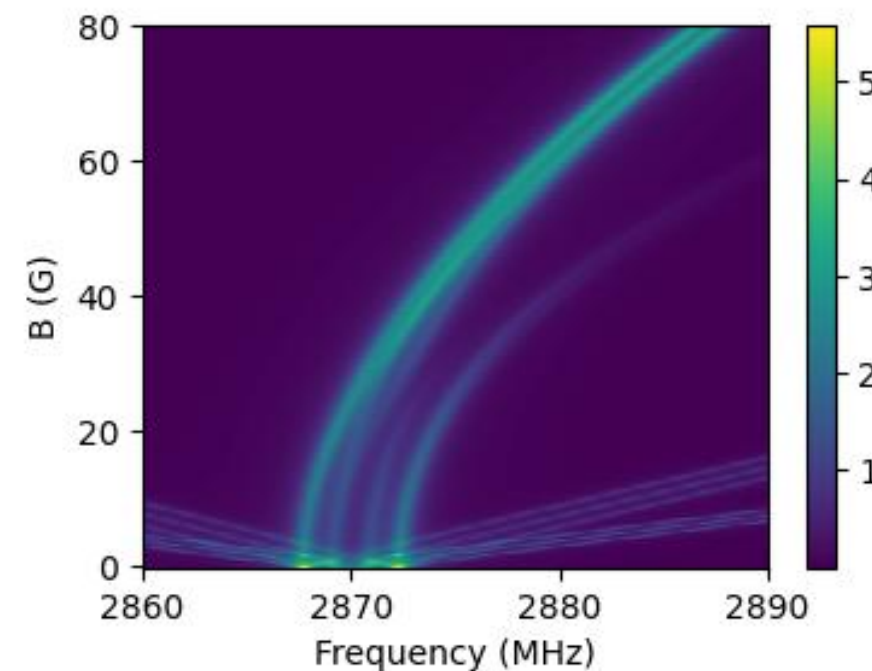


Transition Strength v/s MF

CONCLUSIONS

- ❑ **Low Field ($B=0$):** Several faintly crowded horizontal lines start around 2.87 GHz. These correspond to the ZFH transitions of all 4 NV orientations.
- ❑ **Medium Field ($B<40G$):** Zeeman splitting separates the NV orientations leading to 24 resolved branches.
- ❑ **High Field ($B>40G$):** The inner and outer transitions begin to merge due to their non-zero linewidth.
- ❑ The main bright branch which starts around 2871 MHz at 30G and curves upto 2885 MHz at 80G corresponds to one of the $m_s=0 \rightarrow -1$ transition for an NV family that is most weakly aligned with the MF.

$$4 \text{ orientations} \times 2 \text{ electronic transitions} \times 3 \text{ hyperfine} = 24 \text{ peaks}$$



Transition Strength ν/s MF

ODMR SPECTRA SIMULATION

Differential Electrometry

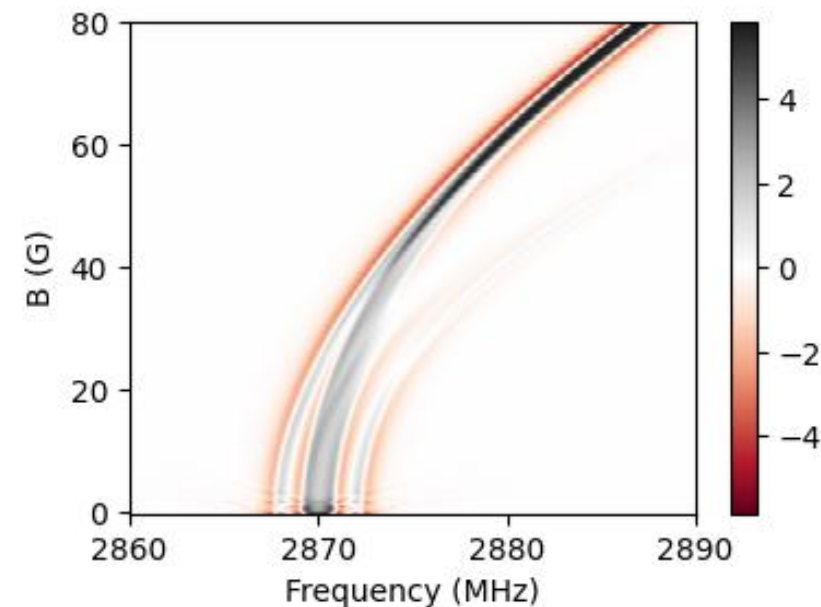
Calculates the electrical sensitivity of NV centre using a differential map

```
fig_S, ax_S = plt.subplots()
sensitivity = (TSmap - TSmap_dE) / dE
scale=1e7
im = plt.pcolormesh(MWfreq, B0list, sensitivity*scale, shading='auto', cmap='RdGy')
plt.colorbar(im)
vmax = max(np.max(sensitivity*scale), abs(np.min(sensitivity*scale)))
im.set_clim(-vmax, vmax)
plt.xlabel('Frequency (MHz)')
plt.ylabel('B (G)')
```

Layout Compression Formatting

```
for fig in [fig_T, fig_S]:
    fig.set_size_inches(4, 3)
    fig.subplots_adjust(bottom=0.17, top=0.92, left=0.14, right=0.95)
```

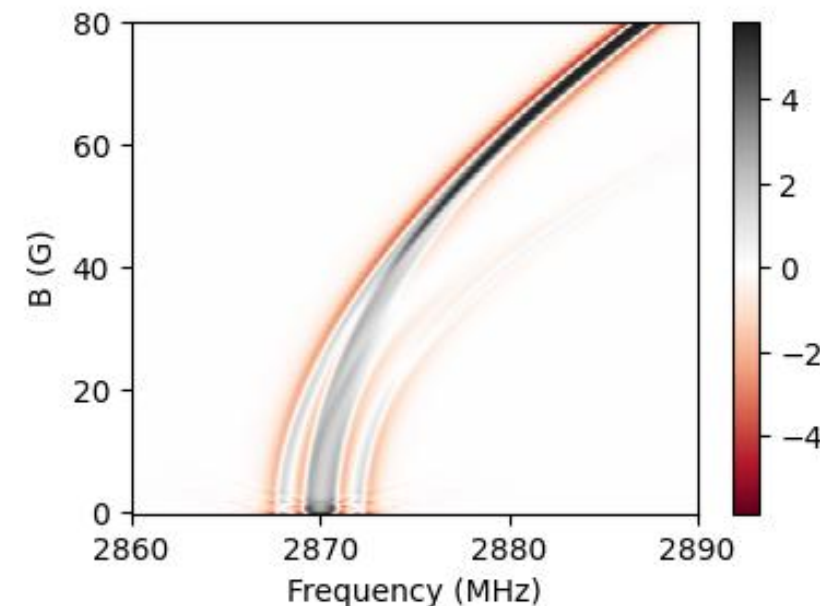
Ensures that labels, ticks and colour bar fit properly without being cut



Electric Field Sensitivity v/s MF

CONCLUSIONS

- ❑ **Sensitivity:** $TS(E_0) - TS(E_0 + dE)$ $E_0 = 5 \times 10^6 \text{ V/m}$
- ❑ **Black (+ve ΔS):** Transition strength decreases as EF increases. $dE = 1 \times 10^5 \text{ V/m}$
- ❑ **Red (-ve ΔS):** Transition strength increases as EF increases.
- ❑ **Low Field (B=0):** All 4 NV orientations are degenerate. Monotonically increasing sensitivity with MF.
- ❑ **Medium Field (B~30G):** The sensitivity doesn't drop to 0 unlike single NV case.
- ❑ **High Field (B~60G):**
 - The Zeeman splitting dominates and the 3 hyperfine lines merge into single unresolved line.
 - Maximal sensitivity is obtained and it saturates.
 - The chosen MW direction allows to selectively excite only the lower frequency transition with TS twice as in unpolarized case.



Electric Field Sensitivity ν/s MF

ODMR SPECTRA SIMULATION

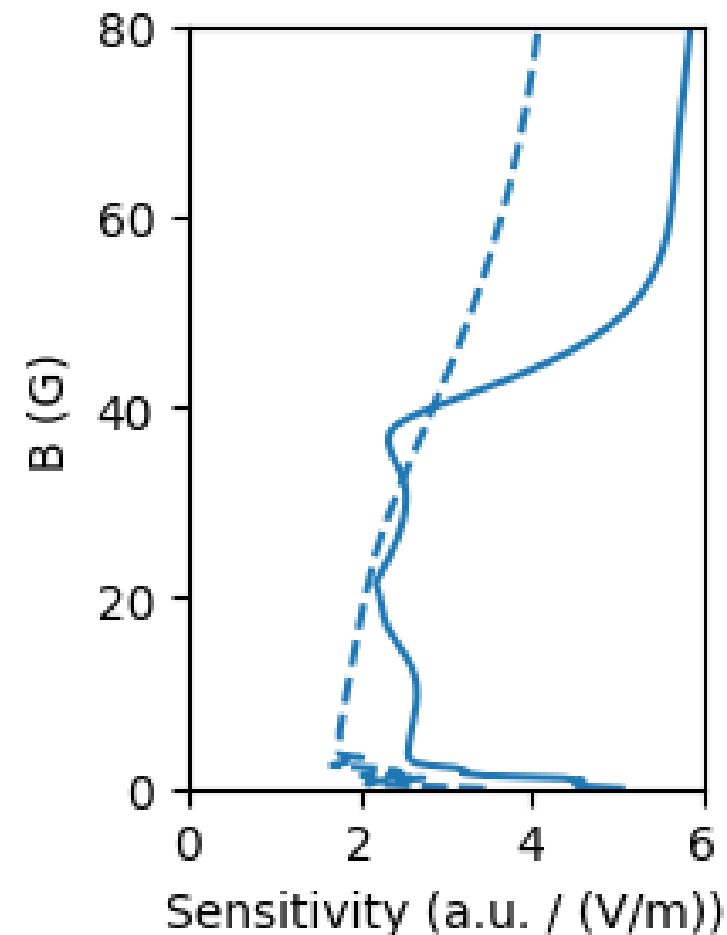
Sensitivity Plot

Shows maximum positive and negative sensitivity as a function of MF

```
fig_max, ax_max = plt.subplots(figsize=(2.2, 3))
sensitivity = (TSmap - TSmap_dE) / dE
plt.subplots_adjust(bottom=0.17, top=0.92, left=0.25, right=0.94)
plt.plot(np.max(sensitivity, axis=-1)*scale, B0list)
plt.plot(-np.min(sensitivity, axis=-1)*scale, B0list, color='C0', ls='--')
plt.ylim(5, B0)
plt.xlim(0,6)
plt.ylabel('B (G)')
plt.xlabel('Sensitivity (a.u. / (V/m))') # Updated units
plt.xticks([0, 2, 4, 6])
plt.yticks([0, 20, 40, 60, 80])
```

Image File Export : Saving the plots if intended

```
if save_flag:
    sname = 'Fig4'
    fig_T.savefig(sdir + sname + '_d.png', dpi=200)
    fig_S.savefig(sdir + sname + '_e.png', dpi=200)
    fig_max.savefig(sdir + sname + '_f.png', dpi=200)
```

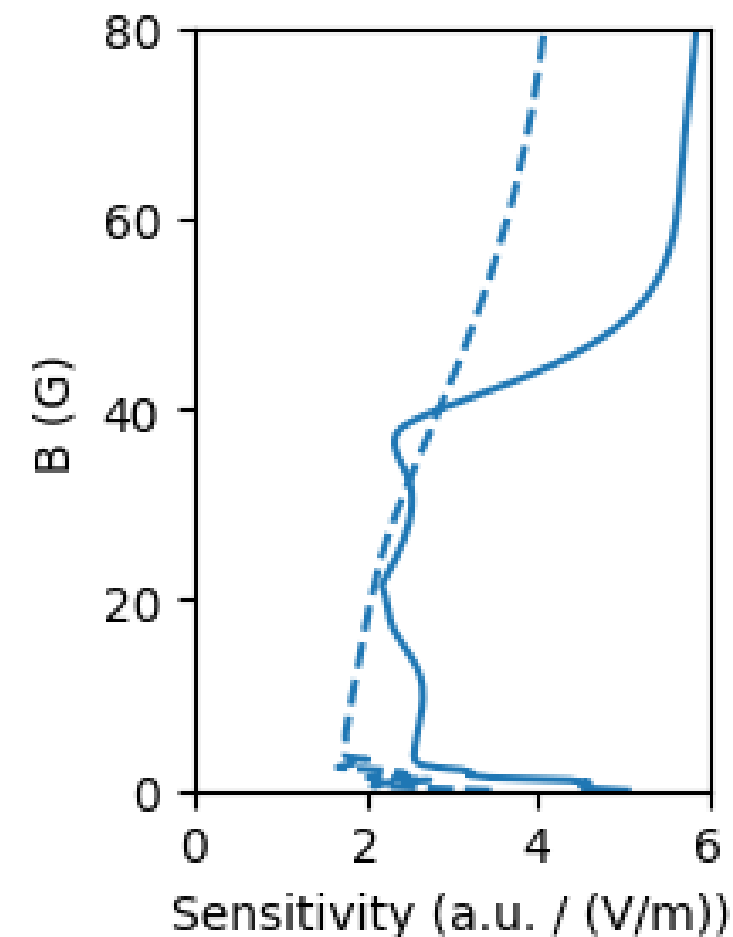


Optimal Values of Sensitivity

CONCLUSIONS

- ❑ **Sensitivity:** $TS(E_0) - TS(E_0 + dE)$
- ❑ **Black (+ve ΔS):** Transition strength decreases as EF increases.
- ❑ **Red (-ve ΔS):** Transition strength increases as EF increases.
- ❑ **Low Field ($B=0$):** Both curves show jagged, noisy behaviour. The sensitivity here is unreliable.
- ❑ **Medium Field ($B < 40G$):** Hyperfine structure is still resolved so 3 distinct hyperfine lines. Sensitivity is diluted because the changes in multiple overlapping transitions partially cancel out.
- ❑ **High Field ($B > 40G$):** The Zeeman splitting dominates and the 3 hyperfine lines merge into single unresolved line. Optimal sensing is achieved at $B=60-80G$ where the contrast is highest.

— : maximum sensitivity
 : minimum sensitivity



Optimal Values of Sensitivity